

WP5.A2 – EcoUnity App Content Architecture and Key Screens

1. Purpose of WP5.A2

WP5.A2 focuses on preparing the content model, educational material prototypes, and reusable templates for the EcoUnity mobile app. The aim is to ensure that all app content is pedagogically consistent, technically manageable, multilingual, inclusive, and ready for later integration into the app.

The app content architecture should support three core learning pillars:

1. **Sustainability education** through the 17 Sustainable Development Goals.
 2. **Social Mixing and inclusion** through collaborative, empathy-building learning activities.
 3. **In-Service Learning** through practical sustainability challenges connected to real-life school and community action.
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2. Proposed App Content Architecture

2.1 Main Content Structure

The app should be organised around **17 SDG learning modules**.

Each SDG module should contain:

1. **Module introduction**
2. **Interactive comic link**
3. **Three micro-learning resources**
4. **Quiz or reflection task**
5. **Practical action challenge**
6. **Progress and badge element**
7. **Teacher-use notes or classroom connection**

This creates the required structure of:

- 17 SDG modules
 - 17 interactive comic links
 - 51 micro-learning resources
 - 17 quizzes or reflection tasks
 - 17 practical action challenges
 - 17 module badges
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3. Learning Flow Per Module

Each SDG module should follow the same student-friendly learning path.

Step 1 – Discover

The learner opens an SDG topic and receives a short introduction.

Content should answer:

- What is this SDG about?
- Why does it matter?
- How does it connect to school, home, friends, and community?
- What can young people do?

Step 2 – Explore

The learner accesses the related interactive comic or scenario.

The comic should introduce a realistic dilemma connected to sustainability and inclusion.

Examples:

- A class must decide how to reduce waste.
- Students from different backgrounds disagree on how to organise a school campaign.
- A community faces a local environmental challenge.

Step 3 – Learn

The learner completes three short micro-learning resources.

Recommended structure:

- **MLR 1 – Understand the issue**
- **MLR 2 – See real-life examples**
- **MLR 3 – Take action**

Step 4 – Reflect

The learner completes a short quiz, decision task, or reflection question.

This should check understanding without feeling like a formal test.

Step 5 – Act

The learner receives a practical challenge.

Examples:

- Create a zero-waste lunch plan.
- Interview a family member about water use.
- Map biodiversity around the school.
- Join a mixed-group classroom challenge.
- Create a poster or mini-campaign.

Step 6 – Progress

The learner earns progress points or a badge after completing the module.

Completion logic:

- Complete all MLRs = module complete
 - Complete quiz/reflection = knowledge check complete
 - Complete action challenge = active citizenship badge progress
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4. Content Types

4.1 SDG Module Page

Each module should include:

- SDG number and title
- Short description
- Visual icon or illustration
- Estimated time
- Learning objectives
- Related comic
- Three MLR cards
- Challenge card
- Badge/progress status

4.2 Micro-Learning Resource Template

Each MLR should be short, visual, and mobile-first.

Recommended length:

- Text: 150–250 words
- Video: 1–3 minutes
- Quiz: 3–5 questions
- Activity: 5–10 minutes

Template fields:

- Title

- SDG link
- Age group
- Learning objective
- Content format
- Key message
- Main content
- Reflection question
- Accessibility notes
- Language version
- Teacher note

4.3 Interactive Comic Link

The app does not need to duplicate the full WP4 repository structure, but each SDG module should clearly connect to the relevant comic.

Comic card fields:

- Comic title
- SDG topic
- Main characters
- Scenario summary
- Decision points
- Inclusion theme
- Sustainability theme
- Start comic button
- Continue comic button

4.4 Quiz Template

Each SDG module should include a short knowledge check.

Question types:

- Multiple choice
- True/false
- Match the idea
- Choose the best action
- Reflection prompt

Avoid overly academic language. Feedback should be encouraging and explanatory.

4.5 Challenge Template

Each module should end with a practical challenge.

Template fields:

- Challenge title
- What to do

- Why it matters
- Individual or group task
- Estimated time
- Materials needed
- Inclusion prompt
- Reflection question
- Completion confirmation

Example inclusion prompt:

Work with someone you do not usually work with. Compare your ideas and agree on one shared action.

5. Proposed Content Categories

The app should classify content using the following tags:

SDG Tags

SDG 1–17.

Learning Type Tags

- Watch
- Read
- Play
- Reflect
- Discuss
- Act
- Quiz
- Challenge

Inclusion Tags

- Teamwork
- Empathy
- Intercultural dialogue
- Anti-discrimination
- Accessibility
- Conflict resolution
- Belonging

Sustainability Tags

- Climate action
- Biodiversity
- Water
- Waste

- Energy
- Responsible consumption
- Food
- Community action

User Level Tags

- Beginner
 - Intermediate
 - Classroom activity
 - Home activity
 - Group challenge
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6. Key Screens of the App

6.1 Welcome / Language Selection Screen

Purpose:

Allow users to choose their language and enter the app easily.

Elements:

- EcoUnity logo
- Language selector
- Short welcome text
- Start button
- Accessibility settings shortcut

Recommended languages:

- English
 - Polish
 - Ukrainian
 - German
 - Finnish
 - Romanian
 - Spanish
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6.2 Onboarding Screens

Purpose:

Explain the app in 3–4 simple screens.

Suggested onboarding flow:

1. Learn about the planet.
2. Explore interactive stories.
3. Complete challenges.
4. Earn badges and take action together.

6.3 Home Dashboard

Purpose:

Provide a simple entry point for students.

Elements:

- Continue learning button
- SDG module grid
- Latest challenge
- Progress summary
- Badge preview
- Teacher/classroom code option, if used

6.4 SDG Module List Screen

Purpose:

Display all 17 SDG modules.

Elements:

- SDG icons
- Module title
- Completion status
- Estimated time
- Badge status

Filters:

- Not started
- In progress
- Completed
- Challenges

6.5 SDG Module Detail Screen

Purpose:

Show the full learning path for one SDG.

Elements:

- SDG title and description
 - Learning objectives
 - Comic card
 - Three MLR cards
 - Quiz/reflection card
 - Challenge card
 - Progress indicator
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6.6 Micro-Learning Resource Screen

Purpose:

Deliver short learning content.

Elements:

- Title
 - Media/content area
 - Key message
 - Interactive element
 - Reflection question
 - Mark as complete button
 - Next activity button
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6.7 Interactive Comic Screen / Comic Launcher

Purpose:

Connect app users to the branching scenario comic.

Elements:

- Comic cover image
 - Scenario summary
 - Characters
 - Start/continue button
 - Related SDG
 - Reflection prompt after completion
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6.8 Quiz / Reflection Screen

Purpose:

Check understanding and encourage reflection.

Elements:

- Question card
 - Answer options
 - Feedback after answer
 - Progress bar
 - Final result
 - Continue to challenge button
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6.9 Challenge Screen

Purpose:

Move learning from the app into real-world action.

Elements:

- Challenge title
 - Step-by-step task
 - Group/individual mode
 - Inclusion prompt
 - Completion checkbox
 - Reflection question
 - Submit/mark complete button
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6.10 Badges and Progress Screen

Purpose:

Motivate learners and show achievements.

Elements:

- Earned badges
- Locked badges
- SDG progress
- Challenge progress
- Final EcoUnity badge progress

Badge categories:

- SDG module badges
 - Challenge badges
 - Teamwork badge
 - Sustainability action badge
 - Final EcoUnity completion badge
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6.11 My Learning Journey Screen

Purpose:

Show the learner's completed modules and next recommended activities.

Elements:

- Completed modules
 - In-progress modules
 - Suggested next module
 - Reflections saved locally or in user profile
 - Total progress
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6.12 Teacher Resources Screen

Purpose:

Support teachers using the app in class.

Elements:

- How to use the app in lessons
- SDG lesson links
- Suggested classroom activities
- Discussion questions
- Challenge facilitation notes
- Accessibility and inclusion tips

This screen can be simple in the first version and expanded later.

6.13 Settings Screen

Purpose:

Allow basic personalisation and accessibility.

Elements:

- Language
 - Text size
 - High contrast mode
 - Audio/subtitle options
 - Privacy information
 - About EcoUnity
 - Partner logos
 - Contact/support
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7. Recommended Navigation Structure

The app should use a simple bottom navigation with four main tabs:

1. **Home**
2. **Learn**
3. **Challenges**
4. **Progress**

Additional areas such as settings, teacher resources, and about page can be accessed from the menu.

8. Content Management Structure

To support multilingual content and reuse of existing Innoventum systems, each content item should be stored as a structured object in the backend.

Recommended object types:

1. SDG Module
2. Micro-Learning Resource
3. Quiz
4. Question
5. Challenge
6. Comic Link
7. Badge
8. Language Version
9. Teacher Note
10. Accessibility Note

Each content object should include:

- Internal ID
- Title
- Short description
- Language
- Related SDG

- Content type
 - Age suitability
 - Status: draft, review, approved, published
 - Partner responsible
 - Last updated date
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9. Prototype Templates to Be Produced in WP5.A2

WP5.A2 should produce the following templates before final content production begins:

1. SDG module template
 2. Micro-learning resource template
 3. Quiz template
 4. Reflection activity template
 5. Practical challenge template
 6. Comic integration template
 7. Badge definition template
 8. Teacher note template
 9. Translation template
 10. Accessibility checklist
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10. Accessibility and Inclusion Guidelines

All content should follow these principles:

- Use simple, age-appropriate language.
 - Avoid long text blocks.
 - Use visual support wherever possible.
 - Provide subtitles or transcripts for videos.
 - Avoid culturally biased examples.
 - Include diverse names, characters, and situations.
 - Avoid competitive ranking between students.
 - Encourage cooperation rather than individual performance only.
 - Make activities possible with low-cost or no-cost materials.
 - Ensure tasks can be completed in different school contexts.
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11. App Content Production Workflow

Recommended workflow:

1. Define SDG module structure.
2. Create prototype for one complete SDG module.

3. Review prototype with teachers and partners.
 4. Adjust templates based on feedback.
 5. Produce remaining SDG module drafts.
 6. Translate and culturally adapt content.
 7. Upload content to backend.
 8. Test content in the app interface.
 9. Revise after pilot feedback.
 10. Approve final content for launch.
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12. Recommended First Prototype Module

The first prototype should be based on **SDG 12 – Responsible Consumption and Production**.

Reason:

It is practical, easy to understand for ages 10–14, and suitable for classroom challenges.

Prototype content package:

- SDG 12 introduction
 - Comic scenario: “The Zero-Waste School Day”
 - MLR 1: What is waste?
 - MLR 2: How can schools reduce waste?
 - MLR 3: What can I do this week?
 - Quiz: 5 questions
 - Challenge: Create a zero-waste classroom plan
 - Badge: Responsible Consumer badge
 - Teacher note: How to run the activity in mixed groups
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13. Minimum Viable App Content for Testing

For the first testable prototype, the app should include:

- Language selector
- Home screen
- SDG module list
- One complete SDG module
- Three MLRs
- One quiz
- One challenge
- One badge
- Basic progress tracking
- Teacher note screen

This will allow partners to test the learning flow before producing all 17 modules.

14. Expected WP5.A2 Outputs

By the end of WP5.A2, the consortium should have:

1. A confirmed app content architecture.
2. A complete list of content types.
3. Reusable templates for all educational materials.
4. One fully drafted prototype SDG module.
5. Draft structure for all 17 modules.
6. Agreed metadata and translation fields.
7. Accessibility and inclusion checklist.
8. Initial screen map for UI/UX development in WP5.A3.